

Syndrome Decoding for a Linear Code

Encoding $E : (\mathbb{Z}/2)^4 \rightarrow (\mathbb{Z}/2)^8$

Generator Matrix and Encoding

We are given the generator matrix

$$G = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 1 \\ 1 & 1 & 1 & 1 & 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 & 1 & 1 & 0 & 1 \end{bmatrix}.$$

The encoding function is

$$E(w) = w \cdot G,$$

where $w = (w_1, w_2, w_3, w_4) \in (\mathbb{Z}/2)^4$ is a row vector and all arithmetic is modulo 2.

An encoded codeword $c = E(w)$ is sent over a noisy channel, and the receiver gets

$$r = 11010110.$$

We assume

$$r = c + e,$$

where e is the (unknown) error pattern and c is the (unknown) transmitted codeword.

Our tasks:

- (a) Compute the syndrome of r .
- (b) Find an error pattern e such that $r = c + e$ with c a codeword.
- (c) Compute the corresponding encoded codeword c .

Parity-check Matrix and Syndrome

For a linear code defined by a generator matrix G , a **parity-check matrix** H is any matrix whose rows span the orthogonal complement of the row space of G (over $\mathbb{Z}_2$). That is, H satisfies

$$GH^T = 0.$$

One valid choice of a parity-check matrix H for this code is:

$$H = \begin{bmatrix} 0 & 0 & 0 & 1 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 0 & 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 & 1 & 1 & 0 & 0 \end{bmatrix},$$

and indeed one can check that $GH^T = 0 \pmod{2}$.

For any received vector r , the **syndrome** is defined as

$$s = Hr^T \in (\mathbb{Z}_2)^4.$$

(a) Compute the Syndrome of r

The received vector is

$$r = (1, 1, 0, 1, 0, 1, 1, 0).$$

We compute

$$s = Hr^T$$

row by row (mod 2).

Let $H = (h_{ij})$ with rows h_1, h_2, h_3, h_4 .

Row 1

$$h_1 = (0, 0, 0, 1, 1, 0, 1, 0)$$

Dot product:

$$h_1 \cdot r^T = 0 \cdot 1 + 0 \cdot 1 + 0 \cdot 0 + 1 \cdot 1 + 1 \cdot 0 + 0 \cdot 1 + 1 \cdot 1 + 0 \cdot 0.$$

Compute:

$$= 1 + 0 + 1 = 2 \equiv 0 \pmod{2}.$$

So the first syndrome bit is 0.

Row 2

$$h_2 = (0, 0, 1, 0, 0, 0, 1, 1)$$

Dot product:

$$h_2 \cdot r^T = 0 \cdot 1 + 0 \cdot 1 + 1 \cdot 0 + 0 \cdot 1 + 0 \cdot 0 + 0 \cdot 1 + 1 \cdot 1 + 1 \cdot 0.$$

Compute:

$$= 0 + 0 + 0 + 0 + 0 + 0 + 1 + 0 = 1.$$

So the second syndrome bit is 1.

Row 3

$$h_3 = (0, 1, 0, 0, 0, 0, 1, 0)$$

Dot product:

$$h_3 \cdot r^T = 0 \cdot 1 + 1 \cdot 1 + 0 \cdot 0 + 0 \cdot 1 + 0 \cdot 0 + 0 \cdot 1 + 1 \cdot 1 + 0 \cdot 0.$$

Compute:

$$= 1 + 1 = 2 \equiv 0 \pmod{2}.$$

So the third syndrome bit is 0.

Row 4

$$h_4 = (1, 0, 0, 0, 1, 1, 0, 0)$$

Dot product:

$$h_4 \cdot r^T = 1 \cdot 1 + 0 \cdot 1 + 0 \cdot 0 + 0 \cdot 1 + 1 \cdot 0 + 1 \cdot 1 + 0 \cdot 1 + 0 \cdot 0.$$

Compute:

$$= 1 + 0 + 0 + 0 + 0 + 1 + 0 + 0 = 2 \equiv 0 \pmod{2}.$$

So the fourth syndrome bit is 0.

Syndrome result

Putting all four bits together:

$$s = \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix} \Rightarrow \boxed{s = 0100.}$$

(b) Compute an Error Pattern e such that $r = c + e$

For any received word r , we have

$$r = c + e$$

with c a codeword ($Hc^T = 0$) and e the error pattern. Applying H :

$$Hr^T = H(c + e)^T = Hc^T + He^T = 0 + He^T = He^T.$$

So the syndrome depends only on e , and

$$s = He^T.$$

We already computed $s = 0100^T$.

A standard fact in syndrome decoding: the syndrome of a *1-bit* error in position j (i.e. $e = e^{(j)}$ with a single 1 in position j) is equal to the j -th **column** of H .

Let $H = [h^{(1)} \ h^{(2)} \ \dots \ h^{(8)}]$ where $h^{(j)}$ is the j -th column. Then if e has a 1 *only* in position j ,

$$e = (0, \dots, 0, 1, 0, \dots, 0), \quad \Rightarrow \quad He^T = h^{(j)}.$$

We want $He^T = s = (0, 1, 0, 0)^T$. So we look for columns of H equal to 0100^T .

Writing the columns of H explicitly:

$$H = \begin{bmatrix} 0 & 0 & 0 & 1 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 0 & 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 & 1 & 1 & 0 & 0 \end{bmatrix} = \left[\begin{array}{c|c|c|c|c|c|c|c} \begin{array}{|c|} \hline h^{(1)} \\ \hline \end{array} & \begin{array}{|c|} \hline h^{(2)} \\ \hline \end{array} & \begin{array}{|c|} \hline h^{(3)} \\ \hline \end{array} & \begin{array}{|c|} \hline h^{(4)} \\ \hline \end{array} & \begin{array}{|c|} \hline h^{(5)} \\ \hline \end{array} & \begin{array}{|c|} \hline h^{(6)} \\ \hline \end{array} & \begin{array}{|c|} \hline h^{(7)} \\ \hline \end{array} & \begin{array}{|c|} \hline h^{(8)} \\ \hline \end{array} \end{array} \right].$$

The columns are:

$$\begin{aligned} h^{(1)} &= (0, 0, 0, 1)^T, \\ h^{(2)} &= (0, 0, 1, 0)^T, \\ h^{(3)} &= (0, 1, 0, 0)^T, \\ h^{(4)} &= (1, 0, 0, 0)^T, \\ h^{(5)} &= (1, 0, 0, 1)^T, \\ h^{(6)} &= (0, 0, 0, 1)^T, \\ h^{(7)} &= (1, 1, 1, 0)^T, \\ h^{(8)} &= (0, 1, 0, 0)^T. \end{aligned}$$

We see that

$$h^{(3)} = (0, 1, 0, 0)^T = s \quad \text{and} \quad h^{(8)} = (0, 1, 0, 0)^T = s.$$

This means that there are at least two possible single-bit error patterns that give the same syndrome:

- An error in position 3: $e = 00100000$,
- or an error in position 8: $e = 00000001$.

Both satisfy $He^T = s$, so both are mathematically valid error patterns consistent with the syndrome. The code is not strong enough to distinguish uniquely between these two single-bit errors.

For concreteness, we choose the error in position 3:

$$\boxed{e = 00100000.}$$

(c) Compute the Encoded Message c

We know:

$$r = c + e \quad \Rightarrow \quad c = r + e$$

with addition modulo 2 (bitwise XOR). Using

$$r = 11010110, \quad e = 00100000,$$

we compute

$$c = 11010110 \oplus 00100000 = 11110110.$$

So:

$$\boxed{c = 11110110.}$$

We can quickly verify that c is indeed a codeword by checking it is equal to one of the rows of G or a linear combination of rows of G .

Note that the third row of G is

$$r_3 = (1, 1, 1, 1, 0, 1, 1, 0) = 11110110,$$

so

$$c = r_3 = E(0010),$$

confirming that c is a valid codeword.

Answer Summary

(a) Syndrome of r :

0100

(b) One valid error pattern (single-bit error in position 3):

00100000

(c) Corresponding encoded message (codeword):

11110110